

KARIN KJELLGREN

Albert-Ueberle-Str. 2, 69120 Heidelberg, Germany, karin.kjellgren01@stud.uni-heidelberg.de

EDUCATION

- 09/2023 – Present Ph.D. in Astrophysics
Heidelberg University
Advisors: Philipp Girichidis, Ralf Klessen
Preliminary thesis title: *The impact of cosmic rays in Milky Way-like galaxies and their gamma-ray signatures*
- 08/2021 – 06/2023 Master's Programme in Astronomy
Stockholm University
Advisors: Sambit Giri, Garrelt Mellema
Thesis: *Modeling the 21-cm bias from the Epoch of Reionization using Machine Learning*
- 04/2022 – 10/2022 Erasmus Student Exchange Programme
Heidelberg University
- 08/2017 – 06/2020 Bachelor's Programme in Astronomy
Stockholm University
Advisors: T. Emil Rivera-Thorsen, Angela Adamo
Thesis: *Physical properties of recombination nebulae in a gravitationally lensed, Lyman continuum emitting starburst at $z=2.37$ from rest-frame optical line emission*
- 08/2014 – 05/2017 International Baccalaureate
Kungsholmens gymnasium
- Bilingual diploma in Swedish and English

FIRST-AUTHOR PUBLICATIONS

The diffuse gamma-ray sky of a Milky Way analog: Local diversity and global constraints, 2026, A&A, 710, A163

The dynamical impact of cosmic rays in the Rhea magnetohydrodynamic simulations, 2025, A&A, 700, A124

INVITED & CONTRIBUTED TALKS

BUGS2026: Building Galaxies from Scratch II, Sep. 7 – 11, 2026, Heidelberg, Germany, **contributed talk**

9th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy, Aug. 31 – Sep. 4, Heidelberg, Germany, **contributed talk**

Workshop on Zoom-in Views of Galaxy Disks Across Cosmic Time: Bridging Simulations and Observations, Jun. 1 – 19, 2026, Stockholm, Sweden, **contributed talks**

Lund Observatory Division of Astrophysics weekly seminar, Apr. 16, Lund, Sweden, **invited talk**

ECOGAL weekly seminar, Oct. 4, 2025, online, **invited talk**

MIST2025 - Cosmic turbulence and Magnetic fields: physics of baryonic matter across time and scales, Cargèse, France, Sep. 29 – Oct. 3, 2025, **contributed talk**

MPIK group seminar, Jun. 3, 2025, Heidelberg, Germany, **invited talk**

Physics of Star Formation: ISM dynamics and star formation, Dec. 3 – 6, 2024, Heidelberg, Germany, **contributed talk**

Cosmic Rays 3, Oct. 22 – 24, 2024, Florence, Italy, **contributed talk**

8th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy, Sep. 2 – 6, 2024, Milan, Italy, **contributed talk**

OTHER SCHOOLS & CONFERENCES

Durham Advanced School on Astrophysical Simulations, Sep. 15 – 19, 2025, Durham University, UK

Magnetic Fields and Cosmic Rays across Diverse Scales: What's Next?, Sep. 8 – 12, 2025, CfA, USA

18th Potsdam Thinkshop, Jul. 14 – 18, 2025, AIP, Germany

ALMABO 2024: View on the multi-phase interstellar medium in galaxies, Sep. 9 – 13, 2024, Bologna, Italy

ECOGAL school on “The Physics of Star Formation”, Feb. 12 – 23, 2024, Les Houches, France

OUTREACH AND OTHER ACADEMIC WORK

09/2026	Member of the LOC <i>9th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy</i>
2025	Teaching assistant <i>Introduction to astronomy II</i>
04/2025	Workshop leader <i>Girls' Day, University of Heidelberg</i>
2024	Teaching assistant <i>Astronomical techniques</i>
09/2022	Volunteer <i>ICAS 2022 Congress, Stockholm</i>
04/2018	Organizer of the “Knut Lundmark”-days <i>The Swedish Astronomical Youth Association</i>
09/2017	Representative at the National Museum of Science and Technology <i>The Swedish Astronomical Youth Association</i>

LANGUAGE SKILLS

Swedish (native), English (professional), French (B2), German (B2)